

UNIVERSITY OF RWANDA

College of Science and Technology

School of ICT

Department of Computer and Software Engineering

WEB DESIGN TECHNICAL REPORT 2025–2026**DIGITAL MINING SAFETY AND MONITORING SYSTEM**

Case Study: Rwanda Mining Industry

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CHAPTER ONE**1.1 Historical Background**

Mining is one of Rwanda's major economic sectors and contributes significantly to employment, foreign exchange earnings, and industrial development. Rwanda is internationally known for producing tin, tantalum (coltan), tungsten, and gold. Although the mining sector has experienced considerable growth, many mining sites continue to rely on traditional artisanal mining methods.

These methods often involve manual excavation, limited monitoring technologies, poor underground ventilation, and inadequate safety equipment. As a result, mining workers are exposed to hazards such as tunnel collapses, gas leaks, flooding, equipment failures, and environmental pollution.

To modernize the sector, the Rwanda Mines, Petroleum and Gas Board (RMB) has introduced initiatives such as the Geological Information and Mining Cadastre System (GIMCS), drone-based mapping, mechanized mineral processing, and environmental monitoring technologies. However, many mining sites still require an integrated digital platform capable of monitoring workers, reporting hazards, and supporting emergency response.

The Digital Mining Safety and Monitoring System was therefore developed to provide a centralized web-based platform that improves worker safety, supports environmental protection, and enhances decision-making through digital technologies.

1.2 Problem Statement

Mining operations in Rwanda continue to experience operational and safety challenges including low mineral recovery, tunnel collapses, poor ventilation, environmental degradation, illegal mining, high operational costs, limited real-time monitoring, slow emergency communication, and lack of centralized digital records.

1.3 Project Objectives

General Objective: Develop a web-based Digital Mining Safety and Monitoring System that improves safety, monitoring, communication, and environmental management.

Specific Objectives: Develop hazard reporting, monitor safety in real time, improve communication, analyze hazards, support emergency response, promote environmental protection, and improve decision-making.

1.4 Expected Outcomes

Reduce workplace accidents, improve worker safety, increase operational efficiency, improve mineral recovery, enhance environmental protection, improve emergency response, support data-driven decisions, and digitize mining safety management.

1.5 Project Rationale

The system provides a centralized platform for hazard reporting, monitoring mining conditions, and supporting informed decision-making.

CHAPTER TWO

System Analysis: Intended users include Mine Workers, Safety Officers, Site Supervisors, Mine Managers, Environmental Officers, and the System Administrator. Intended partners include RMB, Mining Companies, Environmental Agencies, Emergency Response Teams, and Local Communities.

System Design includes System Architecture, Website Navigation Structure, Flowcharts, User Interface Designs, ER Diagram, and Database Tables.

CHAPTER THREE

Implementation Technologies: HTML5, CSS3, JavaScript, Visual Studio Code, and Responsive Design.

Screenshots to include: Home, About, Hazard Reporting, Dashboard, Workers, Alerts, Technology, and Contact pages with captions and explanations.

CHAPTER FOUR

Conclusion

The Digital Mining Safety and Monitoring System addresses key challenges in Rwanda's mining industry by providing a digital platform for safety management, hazard reporting, worker monitoring, and environmental awareness. The website demonstrates how modern web technologies can improve mining safety, enhance communication, and support sustainable mining practices.